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L1F22BSCS0070

QUESTION: 1

i) $\frac{dy}{dx} = \sin 5x$

$dy = \sin 5x\, dx$

$\int dy = \int \sin(5x)\, dx$

$y = -\frac{1}{5}\cos(5x) + c$

2) $\frac{dy}{dx} = (x+1)^2$

$dy = (x+1)^2 dx$

$\int dy = \int (x+1)^2 dx$

$y = \frac{(x+1)^3}{3} + c$

3) $dx + e^{3x} dy = 0$

$e^{3x} dy = -dx$

$dy = -e^{-3x} dx$

Integrate b.s

$\int dy = \int -e^{-3x} dx$

$y = \frac{1}{3} e^{-3x} + C$

4) $dy - (y-1)^2 dx = 0$

$dy = (y-1)^2 dx$

$\frac{dy}{(y-1)^2} = dx$

Integrate B.S

$\int (y-1)^{-2} dy = \int dx$

$-(y-1)^{-1} = x + C$

$\frac{-1}{y-1} = x + C$

$y = 1 - \frac{1}{x+C}$

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5) $x\frac{dy}{dx} = 4y$

$\frac{dy}{y} = \frac{4}{x}dx$

Integrate B.S

$\int \frac{dy}{y} = \int \frac{4}{x}dx$

$\ln|y| = 4\ln|x| + C_1$

Solve for y

$\ln|y| = \ln|x^4| + C_1$

$|y| = e^{\ln|x^4| + C_1}$

$= e^{C_1 - e\ln|x^4|}$

$y = Ax^4$

6) $\frac{dy}{dx} + 2xy^2 = 0$

$\frac{dy}{dx} = -2xy^2$

$\frac{dy}{y^2} = -2x\,dx$

$\int y^{-2}dy = 2\int x\,dx$

$-1/y = -x^2 + C$

$y = 1/x^2 + C$

7) $\frac{dy}{dx} = e^{3x} + 2y$

$\frac{dy}{dx} = e^{3x}e^{2y}$

$\frac{dy}{e^{2y}} = e^{3x}dx$

$e^{-2y}dy = e^{3x}dx$

Integrate b.s

$\int e^{-2y}dy = \int e^{3x}dx$

$= -1/2\, e^{-2y} = \frac{1}{3}e^{3x} + C$

8) $e^{x}y(dy/dx) = e^{-y} + e^{-2x-y}$

$e^{x}y\frac{dy}{dx} = e^{-y}(1 + e^{-2x})$

$e^{x}y\frac{dy}{dx} = 1 + \frac{e^{-2x}}{e^{y}}$

$e^{x}y\,e^{y}\frac{dy}{dx} = 1 + e^{-2x}$

$ye^{x+y}\left(\frac{dy}{dx} = 1 + e^{-2x}\right)$

$ye^{y}dy = \frac{1 + e^{-2x}}{e^{x}}dx$

$ye^{y}dy = (e^{-x} + e^{-3x})dx$

$\int ye^{y}dy = \int e^{-x} + e^{-3x}dx$

$(y-1)e^{y} = -e^{-x} - \frac{1}{3}e^{-3x} + C$

9) $y \ln x \frac{dx}{dy} = \frac{(y+1)^2}{x}$

$$y \ln x \frac{dx}{dy} = \frac{(y+1)^2}{x^2}$$

$$x^2 \cdot \ln x \, dx = \frac{(y+1)^2}{y} dy$$

$$x^2 \ln x \, dx = \left(\frac{y^2+2y+1}{y}\right) dy$$

$$x^2 \ln dx = \left(y + 2 + \frac{1}{y}\right) dy$$

$$\int x^2 \ln x \, dx = \int \left(y + 2 + \frac{1}{y}\right) dy$$

$$\frac{x^3}{3} \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} dx = \frac{y^2}{2} + 2y + m|y| + c$$

$$\frac{x^3}{3} \ln x - \int \frac{x^2}{3} dx = \frac{y^2}{2} + 2y + m|y| + c$$

$$\frac{x^3}{3} \ln |x| - \frac{x^3}{9} = \frac{y^2}{2} + 2y + m|y| + c$$

10). $\frac{dy}{dx} = \left(\frac{2y+3}{4x+5}\right)^2$

$$\frac{dy}{(2y+3)^2} = \frac{dx}{(4x+5)^2}$$

$$(2y+3)^{-2} dy = (4x+5)^{-2} dx$$

$$\int (2y+3)^{-2} dy = \int (4x+5)^{-2} dx$$

$$\frac{1}{2} \frac{(2y+3)^{-1}}{-1} = \frac{1}{4} \left(4x + \frac{5}{-1}\right)^{-1} + C$$

$$\frac{-1}{2(2y+3)} = \frac{-1}{4(4x+5)} + C$$

23) $\frac{dx}{dt} = 4(x^2+1)$, $x(\pi/4) = 1$

$$\int \frac{1}{x^2+1}\,dx = \int 4\,dt$$

$$= 4t + C_1$$

$$= 4\left(\frac{\pi}{4}\right) + C_1$$

$$\frac{\pi}{4} = \pi + C_1$$

$$C_1 = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$$

$$4t - \frac{3\pi}{4}$$

$$x(t) = \tan\left(4t - \frac{3\pi}{4}\right)$$

24) - $\frac{dy}{dx} = \frac{y^2-1}{x^2-1}$, $y(2) = 2$

$$\int \frac{1}{y^2-1}\,dy = \int \frac{1}{x^2-1}\,dx$$

$$\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C_2$$

$$\ln\left|\frac{y-1}{y+1}\right| = \ln\left|\frac{x-1}{x+1}\right| + C_3$$

$$\left|\frac{y-1}{y+1}\right| = e^{C_3}\left|\frac{x-1}{x+1}\right|$$

$$\frac{y-1}{y+1} = C\frac{x-1}{x+1} \qquad \therefore \quad \therefore C = \pm e^{+3}$$

$$\frac{2-1}{2+1} = C\frac{2-1}{2+1}$$

$$\frac{1}{3} = C\,\frac{1}{3}$$

$$C = 1$$

$$\frac{y-1}{y+1} = \frac{x-1}{x+1}$$

$$(y-1)(x+1) = (y+1)(x-1)$$

$$xy + y - x - 1 = xy - y + x - 1$$

$$2y = 2x$$

$$y = x$$

$$y(x) = x$$

(2.3)

i) $\frac{dy}{dx} = 5x$

$\frac{dy}{y} = 5x\,dx$

Integrate b.s

$\ln y = 5x + C$

$y = Ce^{5x}$

2) $\frac{dy}{dx} + 2y = 0$

$\frac{dy}{dx} = -2y$

$\frac{dy}{y} = -2dx$

$\ln y = -2x + C$

$y = Ce^{-2x}$

3) - $\frac{dy}{dx} + y = e^{3x}$

$P(x) = 1,\ f(x) = e^{3x}$

$I(x) = e^{\int 1 dx} = e^{x}$

$e^{x}\frac{dy}{dx} + ye^{x} = e^{3x}e^{x}$

$\frac{d}{dx}\left[ye^{x}\right] = e^{4x}$

$= \frac{1}{4}e^{4x} + C$

$y = \frac{1}{4}e^{3x} + Ce^{-x}$

4) $3\frac{dy}{dx} + 12y = 4$

Divide by 3

$\frac{dy}{dx} + 4y = \frac{4}{3}$

$IF = e^{4x}$

$\frac{d}{dx}\left(ye^{4x}\right) = \frac{4}{3}e^{4x}$

$ye^{4x} = \frac{4}{3}\frac{e^{4x}}{4} + C$

$ye^{4x} = \frac{1}{3}e^{4x} + C$

$y = \frac{1}{3} + Ce^{-4x}$

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6). $y' + 3x^2 y = x^2$

$P(x) = 3x^2 \quad f(x) = x^2$

$I(x) = e^{\int 3x^2 dx}$

$= e^{x^3}$

$\frac{d}{dx}\left[e^{x^3} y\right] = x^2 e^{x^3}$

let $u = x^3$

$du = 3x^2 dx$

$\int x^2 e^{x^3} dx = \frac{1}{3} e^{x^3}$

$y e^{x^3} = \frac{1}{3} e^{x^3} + C$

$y = \frac{1}{3} + Ce^{-x^3}$

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